This Is Your Brain on Psychedelics

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis: Psychological and neurological research led by Ghul Dolan indicates that psychedelic substances do not merely alter consciousness but serve as a biological key to reopening "critical periods"—developmental windows of high neuroplasticity that naturally close in adulthood. By restoring the brain's capacity for structural remodeling (meta-plasticity), psychedelics allow adults to reconfigure memories and behavioral patterns, particularly in treating trauma or motor deficits, provided they are administered within a supportive therapeutic context (setting and setting).

2. Key Arguments & Data:

3. **Critical Period Mechanism:** The concept was originally defined by Conrad Lorenz (1935) regarding imprinting behavior in goslings. Dolan posits that critical periods govern learning capacity across species; failure to close them leads to conditions like autism, while successful closure enables adult stability.
4. **Neuroplasticity Convergence:** All major psychedelics share a downstream unifying mechanism involving the restoration of plasticity (meta-plasticity) and extracellular matrix remodeling, despite binding to different receptors (e.g., LSD targets Serotonin 2A; Ketamine targets NMDA).
5. **Duration Correlation:** The reopening window is proportional to the duration of the subjective trip:
    - Ketamine (30 min–1 hr trip): Critical period remains open ~48 hours.
    - MDMA/Psilocybin (3–6 hr trip): Critical period remains open ~2 weeks.
    - Ibogaine (2–3 day trip): Critical period remains open >1 month (potentially longer).
6. **Species-Specific Evidence:** Research using octopuses (invertebrates lacking a cerebral cortex) with MDMA revealed that the drug reverses asocial behavior into social attachment, proving the molecular

mechanisms for complex behavior are conserved across evolutionary distance.

7. **Therapeutic Limitations & Indications:**
    - **Effective:** PTSD reconfiguration, addiction treatment, motor learning rehabilitation (stroke patients using psilocybin paired with occupational therapy).
    - **Ineffective for Neurodegeneration:** Does not cause new neuron growth; therefore, it is not a cure for Alzheimer's or Parkinson's diseases where neuronal loss persists.
    - **Requirement:** Requires "therapeutic context" to prevent negative outcomes like psychosis; isolation context may trigger the wrong type of plasticity (e.g., raves vs. therapy).
8. **Regulatory Status:** Psychedelics are currently Schedule I substances under the Controlled Substances Act (1970). Organizations like MAPS are lobbying for clinical data to remove this status. Ibogaine is Schedule I but has legal ambiguity in parts of Mexico and offers promise for opioid use disorder despite cardiac risks requiring management.

9. **The "Noetic" Property:** A subjective feeling that the experience reveals deeper reality ("I know it now"). While linked to therapeutic benefit, misinterpreting this as absolute knowledge can lead to narcissism or messiah complexes.

10. Actionable Takeaways:

11. **Investigate Clinical Context:** If considering psychedelic therapy for trauma or learning deficits, ensure administration is paired with traditional psychotherapy and occupational therapy rather than used as a standalone experience.
12. **Manage Expectations Regarding Longevity:** Do not assume psychedelics can reverse neurodegenerative disease progression; they are designed to reconfigure existing neural pathways (like PTSD memories) rather than regenerate lost neurons (like in Alzheimer's).
13. **Advocate for Policy Reform:** Given the Schedule I classification, which historically relied on biased social determinants (e.g., marijuana scheduling linked to Black community usage), support legislation

reviews regarding opioid crisis funding and research grants for legal psychedelic pathways.

14. **Understand "Critical Period" Risks:** Recognize that reopening these windows requires stability; opening them without the appropriate environmental or cognitive safety net can result in reinforcing maladaptive behaviors rather than therapeutic ones.

Full Transcript

Dang, octopuses on psychedelics. Can't top that. We'll try. Ah, yes you can. Two octopuses on psychedelics. Coming up, next wave psychedelics on Star Talk. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This. is Star Talk's special edition, which means I got to my right. Right here, Gary O'Reilly. Hi, Gary. Hi, Neil. All right. Yeah. And who else we got here? Who could you be? Gary's younger brother. It's true he is. Chuck, nice. That's right. All right. So we on special edition specialize in the human condition. Yeah. So... I think you put something together today, you and your producers. Yeah. Laying over in LA. In our LA office. And I think this one puts us over the top. Really? Yeah. Over the top of what? I don't know. We'll find out. Anyway, it's about time I think we had this show. That's what I'm saying. That's what I'm saying. The special tradition, as you've been saying, is the human condition. But what happens if we give it a nudge? What if I told you...

we can take back our brains to a time when it was eager to learn the lessons that would last a lifetime. Go back to last Tuesday? If you wish. I mean, spoiler, Chuck, there are no shamans in this episode. Oh, come on, man. Well, so you don't mean evolutionarily. You mean just in our own lifetime. Yeah. OK. You're not going to go back to our lizard brain. You're just going to go back to our childhood brain. Yes. OK. Unless you're a lizard. That was good. That was good. All right, Neil, please bring it on. We have, for the first time on Start Talk, Ghul Dolan. Ghul, welcome to Start Talk. Thank you very much. Nice to be here. You are professor at UC Berkeley in the Department of Psychology.

But you're also attached to the departments of psychedelics. All right, now we're talking. And neuroscience. Yeah. Wait, Berkeley has a department of psychedelics. Is that true? No, no, no. I'm neuroscience and psychology are the two departments I'm associated with. And then we have a psychedelic center. And I'm part of that as well. Oh, I wish I was a part of the psychedelic center. Stop. And a co-author of a book from 10 years ago. the essential pathways and circuits of autism. Ooh, interesting. That's

something that, oh man. We should have had her on for that. Yeah, yeah, why would we have you on? Yeah, that's a great conversation. We'll get you back. We've got to have you back for that. For autism, for sure. A leader in the field of psychedelics I have here unlocking our neuroplasticity. Yeah. One of my favorite words of the last decade or so. That one along with terraforming.

that came out of astrophysics, words that just no other word was there. And we have it here, neuroplasticity and for the treatment of PTSD, something that has been long sought after of course. So, did you come to this field as just an ordinary psychologist or did you have pure neuroscience psychedelic chops or at least curiosities before you became an academic? Yeah, I've been studying the brain since I don't know 1998 and I started out first studying wholesale patch clamp electrophysiology doing learning and memory studies and Then I switched over to doing in vivo electrophysiology and studying critical periods, which I'll tell you about as they relate to diseases like autism And then I finished medical school actually and I did an MD-PhD and then I decided I didn't really want to see patients and then I wanted to keep doing research. So then I did a postdoc at Stanford and there it was also neuroscience focused on discovering the neural circuit underlying social reward learning and then In 2014, I started my own lab and then I finally got to work on the stuff that I really wanted to work on, including psychedelics and octopuses. What an underachiever. I love how you just threw that in as a throwaway. Psychedelics and octopuses. They go together perfectly.

Of course, they go together. Every time I get in the ocean, I'm like, I hope I see an octopus who's tripping his tentacles off. Well, I think Ringo's star beat us to it on the octopus's garden business. All right. Oh, okay. That's right. So could you explain what the human fascination has been for millennia, for things that alter our state of awareness? And I ask that. very sincerely, because as a astrophysicist, I highly value objective reality. And so to alter my state of mind for whatever tripping that involves, I so undervalue that compared with objective reality. And so can you help me understand people's urge to escape reality? You know, I'm not sure that it was originally necessarily discovered by humans. I think that there's evidence that animals use psychoactive drugs, and we sort of tried them out,

testing them on ourselves, watching the animals do it first. And in fact, there's a long line of evidence to suggest that psychedelics were used in traditional cultures as medicines more than just for the fun of it. Makes sense. And we learn how to use plants and fungi from...

animals probably, mostly in the healing capacity more than the altered states of consciousness kind of capacity. However, I will say that we have some mechanistic insight into what psychedelics do to the brain, and I'll talk about it more, but we think that they reopen critical periods. And it turns out that psychedelics aren't the only way to do that, and humans are have been doing those other practices for a really long time as well. So what you're saying is psychedelics worked its way into cultures with tap roots in medicine. And now we're making coming full circle looking to psychedelics once again to see how it can serve people who are in desperate need. Yes. Listen, the tribe that I did ayahuasca with in Costa Rica, they are from Columbia.

they have been practicing this before recorded time. These same lineage of people. And they do not call it ayahuasca. They call it the medicine. Ayahuasca is our word. They call it the medicine. And that's how they refer to it. And the ceremonies and everything that they do in the ceremonies are geared towards healing. And so that's the way they see it. All right. Interesting. So LSD was formulated as a respiratory and circulatory treatment in 1938, if I'm not mistaken. So when did our perception of what psychedelics could actually do change? Well, you know, Albert Hoffman, who was the first to synthesize LSD, is famous for having decided it must be having some big... psychotropic effect, though he can't really pinpoint how he got that notion in his head, maybe in a dream. He just had an intuition that it was doing something more than what he could tell it was happening when he gave it to a mouse. So he estimated how much he should take based on the potency of another drug like aspirin or ibuprofen. And based on that estimate, he took approximately the equivalent of 250 hits of acid and road is bicycle home.

So LSD is one of the most potent molecules that we know of. And he had, it was the first time anybody had taken this molecule, any human had taken this molecule. And he had a very, very big experience. He was accompanied

by his colleague and a nurse who helped him go through that experience. And then he realized that it was a really different kind of medicine and that it was potentially useful at the time when they first started researching it, useful as a way of trying to understand insanity. So the first name of psychedelics was psychotomimetic because it mimics psychosis is what they thought at the time. Of course, now we know that the psychosis that we see in schizophrenia or other types of drug overdose is very different than the psychedelic properties of psychedelics like LSD. I want to just revisit something you said about critical periods and I'll ask you to explain what they are, but they fold nicely into your work with, is it octopi or octopuses? It's octopuses because it comes from Greek, not Latin, so the Latin is pull-pull. Thank you.

You were working with octopuses with MDMA and you've sort of uncovered some things here. So would you sort of expand on your research and weave into that the explanation of critical periods? When you say working with octopuses and MDMA, were you getting octopuses high? I actually, we were already working with MDMA to study critical periods in mice. And I had become interested in octopuses because another friend of mine, Kerry Alberton, had just published the genome of the first octopus. And I thought, this is great. I'm going to be able to study brain evolution using these species. And a collaborator friend of mine, Eric, said, you know, we have seven octopuses available. Would you be interested in doing any experiments on them? And I said, sure. So he fedexed them to my lab in Baltimore. And then he flew down. And then we gave them MDMA. And we asked what happens to their social behavior.

So octopuses are crazy asocial, so they will kill each other normally if you put two of them in the same tank. Wow, nice. You know, if you give them MDMA, then their behavior completely changes. And so now, instead of... I don't know why we had a problem in the first place, man. I love you, man. I love you. I love you too, man. I'm gonna give you like an eight-arm hug right now. So, the social reward critical period. Okay, I can say it, but... What actually does that mean? Is that one of a whole load of different critical periods that we will have in existence in our brains? Yes. So just to be clear, the octopus experiment was looking at the acute effects of MDMA, and we were not measuring critical periods there. I was going to

say, because from my own experiments with MDMA, surely in the name of science. Of course.

I understand it to be, you know, a serotonin reuptake inhibitor, right? So it's just basically your body, your brain floods itself with a bunch of serotonin and then you feel so good and loving and all the stuff that that does for you. So how close are their brains to ours that they would get that same response? Yeah. So first of all, the MDMA binds to the same transporter protein that a selective serotonin reuptake inhibitor like Prozac binds to, but it's the same transporter. But when an SSRI like Prozac binds to it, it blocks it so that rather than vacuuming up all of the excess serotonin from the synapse, when MDMA binds to it, it reverses the direction. So now it's just extruding a bunch of serotonin into the synapse. So SSRIs work by allowing the excess serotonin to not get vacuumed up. MDMA is much more powerful because it causes actually serotonin to be released, right? Right, yeah. And that's why you feel like shit the next day.

Well, you know, some people report that they feel bad the next day, but other people do not. And so this sort of hangover of social good feeling, that part of it, we haven't really got a big handle on. We know that if you give 50 times the normal recreational dose of MDMA many days in a row to an animal, you can deplete serotonin. from the transporter stores. But it's not clear that that is what's happening when people report that they're having a hangover and it probably has more to do with metabolism of the molecules by the liver. Here's yet another fact from Lost in Space. 5,000 facts for navigating the universe. If Jupiter were hollow, you could fit a thousand Earths inside of it.

And you know what else? Jupiter rotates so fast, it makes one full revolution every 10 hours. If you like that fact, you can find 4,999 more in Lost in Space. 5,000 facts to help navigate the universe. Latest collaboration between StarTalk and National Geographic Books. Lost in Space is now available for pre-order wherever books are sold. Is every psychedelic using the same mechanism to achieve what it needs to achieve or can achieve? Yeah, so Gary, this is a great question. I think that There is a common mechanism across all of the psychedelics, but the proximal mechanism is probably very different across psychedelics. So what I mean

by proximal mechanism is that different psychedelics bind to different receptors at the beginning. So ketamine seems to be working mostly through the NMDA glutamate receptor, whereas psilocybin and LSD They bind to, you know, almost every G protein-coupled receptor we know of, but the one that is responsible for the hallucinogenic properties seems to be the serotonin 2A receptor. MDMA, as we already talked about, binds to the serotonin transporter, but also to the norepinephrine transporter and the dopamine transporter. Whereas Ibogaine is a really sort of unique psychedelic in that it seems to bind a little bit.

and at low affinity to lots of different things, including the Kappa opioid receptor, the NMDA receptor, the serotonin transporter, and not at all at the serotonin 2A receptor. So despite the fact that they bind to different receptors, what we have shown is that downstream of that receptor binding, they all seem to do something similar in terms of restoring something that I'll explain is called meta plasticity. So restoring the ability of the brain to induce plasticity, and they seem to converge at the level of extracellular matrix remodeling. So those mechanisms seem to be the downstream unifying mechanism that brings them all together. You're saying the brain needs to have a plasticity enough to Have a plasticity Yeah, so let me explain why I actually don't like the word neuroplastigen or psychoplastigen Plasticity is something that we that happens all the time like right now I am having you know this wonderful conversation with you guys It's emotionally very strong for me I get to be on this great podcast and so I'm gonna remember a lot of things about this day, probably the hours and days before and after this, because it's a very powerful, emotionally powerful moment for me. Now we feel the same about you, thanks. So, you know, it's going to cause plasticity in my brain.

So everything that you learn is gonna induce some amount of plasticity. But that plasticity is, follows a bunch of rules. It's associative, it depends on how, what else is happening in your life. Most people are able to remember, for example, other big emotionally. salient moments in their lives like the birth of a child, their wedding day, the explosion of the Challenger, the 9-11. Those are days that, you know, get encoded very strongly because they're emotionally salient. And that is all happening by plasticity. We also know that drugs of abuse, like cocaine, induce massive amounts of

plasticity. And the plasticity they induce is sort of not following the rules. It sort of doesn't care whether you're hungry or not hungry. You know, you kind of, when you start using cocaine, it's like, I love it here. I love it there. I love it everywhere. I don't care if I'm hungry for cocaine or not. I just want it, right? And so that type of plasticity or hyperplasticity is associated with addiction. That kind of plasticity is very different than the kind of plasticity that we are observing with psychedelics. What psychedelics do instead is they seem to not...

do so much plasticity by themselves, although of course the trip itself, if you saw God or something, that's probably going to be a very emotionally meaningful memory in and of itself. But the therapeutic effects we think are because what happens after that big you know, psychedelic experience is it restores the ability to induce plasticity to levels that are similar to the ability to induce plasticity when you were young, when those critical periods were open. So, do the effects of this reopening of a critical period last beyond the pharmacological effect of the drug itself? Yes, yes. And so this is actually... Is that the magic? That's that there is some magic there so you know this is something really cool that we discovered when we looked at the time course of critical period reopening when we looked at different psychedelics where the trip last different amounts of time we saw that how long the critical period stays open.

after the acute effects of the psychedelics have worn off is proportional to how long the trip is, right? So if you've got a psychedelic like ketamine where the trip lasts only about 30 minutes to an hour, then the critical period only stays open for about 48 hours and by one week it's closed again. LSD, you know, the trip lasts about eight to 10 hours and the critical period closes at about three weeks. MDMA and psilocybin or something in the middle the trip lasts three to six hours and then the critical period stays open for about two weeks and then it's closed at three weeks I be gang which is the rock star of the psychedelics the trip lasts, you know two to three days and The critical period stays open for at least a month although probably longer than that. We just didn't look any further out Now, yeah, just for clarification, because I wanna make sure I have it, and maybe somebody else is listening. When you say plasticity, what you were talking about, yes or no, is this what you're talking about? You have experiences or behaviors,

occasions, whatever, and they cause, your neurons to fire in a certain way that wires your brain according to the experience that you are having. And then that makes an imprint on you. And that can change based on different experiences and so forth. Is that what we're talking about when you say plasticity? Sorry, yeah, I should have clarified. When I say plasticity, I am talking about synaptic plasticity.

Which is a change in the synaptic weights that are encoded when two neurons talk to each other The weight of those synaptic connections can change they can change in either direction and the pattern of synaptic weights across a neuron, across distributed networks of neurons, changes as the representation of the memory changes. So an analogy. So neurons that wire together, neurons that fire together wire together. Right. And neurons that are out of sync lose their link. That's sort of that. Lose their link. Okay. And so that bi-directional synaptic plasticity is the leading theory of how memories get encoded by the brain. Ah, that's, there you go. That's the key. Why is that still unknown for sure? Well, I've been looking at a brain for like a hundred years. Well, because it's very, very sparsely encoded and it's dynamic and it is distributed across, you know, billions of neurons and trillions of synapses. And so, And it's not encoded in like, philosophers have speculated about the existence of a grandmother cell, for example, like you can just poke a electrode down in the brain and identify the neuron that responds to grandma. We have never found anything that looks like that. Instead, it looks like the way that the brain encodes memories is by population coding. So the changes in the patterns of activity across many, many, many, many neurons is how the memory gets encoded. So an analogy that might be useful is think of a QR code. So if a QR code represents all of the...

cell, a neuron, and the black and white pixels on that QR code each represent a synapse. The memory isn't the whole block turning, you know, more black or more white. It is the pattern of black and white pixels on that QR code. And so the memory is just how the pattern changes, not some sort of gradation, graded effect, which makes it very difficult to measure from a distance. Yeah. And could that also be, and maybe there's a different reason, but what I'm hearing, maybe that's why memories are so unreliable, you know, in terms of the recall. Well, I mean, I think that they can be unreliable

if you think of them as being constantly modified, right? So Things that are very reliable memories are the ones that are essentially encoded because the same event triggered sort of redundant.

Networks reinforce reinforce either reinforce because of a reward signal or reinforce because of a fear signal or reinforce because. You it was so powerful at the moment you your your olfactory system your visual system your emotional system your social system all of these networks encoded it redundantly and so you have lots of different ways to access that memory sometime in the future. Whereas if something just happened today and it was like, oh hum, you know, just another hard boiled egg for breakfast, you probably will only encode that in the least redundant way possible and it will kind of fall away over time. Wow, that's a great explanation. Our neuroscience is now looking at psychedelics as the master key that will unlock the brain's plasticity. Yeah, so that's definitely the way that my lab and like a handful of other labs that have reproduced our findings are our finding. And I just want to back up to you to explain what a critical period is and why this is such a big deal for neuroscience. So critical periods were first discovered and or first described in 1935 by Conrad Lorenz. He used that term to describe imprinting behavior in geese. And what he noticed is is that those baby gosslings sort of within 48 hours after hatching will form a lasting, really robust, learned attachment to whatever is moving around in their environment. So normally that would be their mom, but if mom isn't available it can be any other caregiver in the immediate environment, but it can also be a kooky scientist like Conrad Lorenz.

But after that first 48 hours is over, then they no longer are sensitive to what's moving around in their environment. And they don't learn those attachments. And so Conrad Lorenz called that a critical period. memory that we, every part of learning and memory that we think is important for enabling humans and other organisms to develop complex behaviors, we think now is regulated by a critical period because there aren't enough genes in the genome to encode every possible behavior. Instead, what's encoded is the ability to flexibly learn from your environment depending on what you're growing up with, right? So it doesn't make any sense to have, you know, a genetically hardwired one single language to rule them all, if you

will. Instead, what you're born with is the ability to learn whatever language you happen to hear in your environment when you grew up.

So for me, it doesn't make any sense to be genetically hardwired with Turkish. I learned English because I was born in the United States and then also Turkish because my parents spoke it to me. So language is also governed by a critical period. And what we have discovered over, you know, since 1935 is that there are dozens of critical periods. We haven't discovered all of them yet. And probably the reason that we're so terrible at curing diseases of the brain is because by the time we get around to intervening, the relevant critical period has closed. So neuroscientists have been dreaming of a way to reopen critical periods basically since 1935. And so when I was a graduate student at MIT, I remember there was another competing lab at Harvard and they were making the claim that, you know, we just have to understand these mechanisms and eventually we'll get a master key and we'll be able to unlock them. And I remember sort of making fun of that idea. I was like, no way anything that would do that to the brain would either cause it to structurally melt down or cause amnesia or cause some kind of seizure activity because of what we know about the mechanisms of critical periods. So it was quite surprising to me when we first discovered that psychedelics reopen critical periods and that, in fact, all of them do it.

And now it seems that they share the same mechanisms as other ways of reopening critical periods that we've known about for a really long time. And now a bunch of other labs have shown that they can reopen not just the critical period we discovered, but other critical periods that people have been working on for a really long time. So can I ask you this, okay, based on what you just explained to us, which is, I mean, sounds really cool. There's a lot of potential there of what you just said, but what came to my mind, and I'm a huge advocate for people to use psychedelics for positive purposes, but the way you say this opening a critical period, I also kind of hear maybe opening a Pandora's box.

because that critical period, like maybe there are some things in there that were neatly tucked away and then you open it up and now I'm totally effed because there's things in this critical period are open again to me. So what

do you do for that? So is it a paradigm in your field to presume that access to early states of mind and emotions is always a good thing? Yeah, what he said. Yeah Like right on I love this. Yes, so it's the case that we think that critical periods need to close right like You know and that if you don't close them properly, you can end up with diseases of the brain We think for example autism might represent a failure to close critical periods properly Furthermore, we think that that's because if you've ever walked around like a forest with a two-year-old, you know that it is very difficult. They notice everything. They're curious about everything. They are solving problems in a totally different way than we as adults are solving problems. But it's terribly inefficient. You usually don't get very far. And if there was a saber-toothed tiger running around, you'd be, if everybody was experiencing the way that a child does, you'd be done. And so they need to close. It's less efficient. We need to eventually get into the mode of experiencing the world, relying on our habit-based memories and our learned efficiencies for accomplishing different things. But if, for example, we learned something and it was adaptive at the time, but no longer adaptive. For example, let's say that you were a child and you experienced some horrible trauma, then you probably learned how to, in that moment, adapt to that situation in a very guarded learning to be hyper-vigilant about threats around you and not trusting people. And that was protective and adaptive at the time. But as you get older, and that threat disappears and you're now you know, physically mature and able to defend yourself better, then those kinds of learned responses around your trauma are no longer appropriate. So what we're imagining is that the way that these psychedelics are helping for diseases like PTSD is that they are enabling a sort of reopening of that window of time when you learned so that you can reconfigure the memories and patterns of behavior in a way that is appropriate for your current circumstances. But is that in tandem with the therapeutic phase of maybe some therapy, some psychiatry, and not just to stand alone, just give them psychedelics, they'll be fine. What are the accompaniments that go along with it?

Yeah, so this is another really cool thing about our mechanistic explanation for how psychedelics work is that we have known for decades that and actually Timothy Leary is one of the first psychologists who described it this way. He said that psychedelics effects are dependent on the set and

setting and what he meant by that is the mindset that you're in when you take them and the setting that you're in who else is around you when you take them and that can have a huge effect on whether you have a good trip or a bad trip whether or not you have a therapeutic effect or whether you have something like you know you hang out with Charles Manson and you end up you know murdering people because you think you're accomplishing something great, right?

So that context dependence of psychedelics effects has been known for a really long time. And what we discovered is that critical period reopening with psychedelics is also context dependent. So if we give MDMA, for example, in a social context, we can reopen the social reward learning critical period in mice. But if we give it in an isolation context, we don't open that critical period. And what I think that's analogous to is if you take MDMA and you go to a rave, maybe you're reopening a critical period for socializing or dancing, but it's not going to be the right kind of environment for unlocking those traumatic memories and trying to have a therapeutic effect, which we think requires a therapeutic context in order to achieve.

Wow, that's great. Let me just ask you just as an aside. What is it about ketamine that causes dumbass billionaires to think they know anything about running a government? You know, I'm not gonna answer that. I don't know said dumbass billionaires. I just what I can say is is that in general psychedelics have this ability that this property called the noetic property, which was actually described by William James in the late 1800s. And that noetic property means that when you take a noetic property is that that feeling that, you know, what you're experiencing is the really real. Everything else up until this moment. was sort of a limited view of the world. But now that I've had this experience, now I really, really know. And that noetic property is part of, I think we think it's important for achieving the therapeutic effects of psychedelics. It's also probably appropriate when you're a child to have this kind of approach to everything that you learn, like, this is it, I know it now, and I can really lock in this memory. But it's also, potentially if it's misused or misunderstood or if you weren't aware of the seduction of that kind of, I really know now, you can end up believing that you understand or know more than you do, which is why I think that we have to be sort of careful about the potential for using these compounds

and then ending up with sort of narcissism or potentially sort of a messiah complex. But Apollo 14 astronaut Edgar Mitchell upon returning to Earth later on would create the Noetic Institute. Oh, what? That's right. Here's what happened. On his way back from the moon. Okay. Okay. He was positioned in the capsule in such a way that the rotation of the capsule was in the same plane as the plane of the solar system. So that as the capsule turned, he would see in sequence the Earth, the moon, other planets, the sun, and that would repeat the entire three days back to Earth. And so this experience, he kept thinking to himself, am I getting a deeper understanding of reality by this repeated exposure to something no one ever would otherwise get to see. And so he was certain afterwards or at least confident that there was some state of mind beyond what our normal reality was that might be achievable when you're exposed in this way as he was.

That was his own eyeball game. The solar system was his eyeball game. He was rotating like, I'm high as hell right now. Yes, essentially. Yes, that was his transcendent state of mind. That was the first time I'd heard the word or seen the term no attic. I didn't know what it meant. And you're reaffirming this belief that you have achieved something that no one else has, so therefore we all must come to you for the answers. And it's Edgar Mitchell's quote that I opened one of my books with, which is, you develop an instant global consciousness, a people orientation, an intense dissatisfaction with the state of the world, and a compulsion to do something about it.

From out there on the moon, international politics looks so petty. You want to grab a politician by the scruff of the neck and drag him a quarter million miles out and say, look at that, you son of a bitch. He got hungry, didn't he? Earth floating in the air. Wow. Yeah. That's the same guy who started the Noetic Institute. Yeah. Just saying. And listen, I'm with him. That guy's awesome. That's pretty, that's... He's feeling it. He's feeling it. That is intense, but deservedly so. So when we reopen these critical periods, can they be beneficial for resetting, re-educating, giving us a better look at maybe the sciences, maybe astrophysics, the universe? I'm not high, I'm just trying to get better at calculus. Officer. Officer? Yeah, absolutely, absolutely. This critical period idea certainly extends beyond the notion that it only could be useful for therapeutic.

uh properties but you know also for improving our ability to learn new things when we're adults right so once our critical periods have closed you know the adage you can't teach an old dog new tricks you know the idea here is that if you take psychedelics well maybe you can and so yeah if you had something that Gosh, you know, I, for example, I struggle with learning languages if I wanted, you know, in theory, the idea, if this turns out to be true, is that, you know, we would be able to use this for all types of learning, not just learning around neuropsychiatric illness like PTSD. And in fact, actually, you know, One of the ways that we are testing this idea of the generalizability across lots of different critical periods is that in collaboration with my colleagues at Johns Hopkins, Steve Zeiler, and the team of neurologists and occupational therapists that we are working with, we are using psychedelics, in this case psilocybin, to try and reopen a motor.

Learning critical period for people who had stroke and so after you have a stroke your Motor learning critical period opens for about a month or two months and then it closes and so whatever physical therapy whatever occupational therapy You get within that first two months after you have a stroke. That's kind of it like Whatever improvements you get after that time, that's it but for a lot of people it takes you know that after that is closed, those people still have motor deficits. And so the dream here is that we could go back a year out from the stroke when they're no longer able to get any motor improvements anymore and give psilocybin, pair it this time instead of with psychotherapy, with physical therapy, with occupational therapy. And now the therapeutic context is motor learning. And can we get you know improvements in motor function and rehabilitation. You're taking care of the mind and body. So now you've made me think, can you replicate this sort of practice with dementia patients with Alzheimer's sufferers or is that just one step too far right now? Well right now what it looks like is that unfortunately this mechanism doesn't really suggest a way that it would be useful for Alzheimer's or Parkinson's or any of the neurodegenerative diseases because it doesn't cause neurons to regrow. It doesn't cause neurons to be born. And those neurodegenerative processes are unlikely to be interrupted. So even if you could use psychedelics to get some kind of, you know, motor therapy to help Parkinson's patients, you know, regain some of the functions that they lost by teaching them alternative pathways for, let's

say swallowing or moving, the neurodegenerative process is gonna continue. And so we don't think that they will be that therapeutically useful in those cases. I just have one question about the, when you're talking about learning things or when you open a window, So I read that I forget the particular period, but cocaine was used in marriage therapy, you know, in couples therapy, which not a good idea. However, could you use these psychedelics to reopen a period where the chemical attraction between two people was greatest and kind of spark a reignition of their romance so that they're more open to, you know, receiving one another. You're asking for a friend. Yes, of course. No, I'm not because my wife hates me. Like, let's get it. Let's get it. Don't get it twisted. No, yeah.

Yeah, well, I mean, Alexander Shulgin, who is the chemist who is most frequently associated and who he did the first sort of scientific characterization of MDMA, but also invented hundreds of other psychedelics. He loved MDMA and was particularly enamored of the idea that it could be used exactly for that for couples therapy. And so I think, and his wife, Anne Shulgin was therapist. And in fact, before MDMA became a Schedule 1 drug, it was actually used legally by psychotherapists for helping people be able to explore conflict in a very open and honest and vulnerable way that was especially useful for people who were having difficulty communicating with each other, for example, in couples. Not so much as a, you know, aphrodisiac, but more as a way of being vulnerable and open to get past conflicts that are ordinarily too scary or too, you know, that you might not say because you'd be afraid of hurting somebody's feelings, but to be able to be open and vulnerable in this way seems to be one of the... Well, I'm a sensitive guy, so vulnerability is an aphrodisiac to me. Absolutely. I think that's right. That's right. No, that's right. I think that, you know, people...

who focus on just sort of the immediate attraction rather than the importance of these kinds of more vulnerable states for building long-term trust and being able to really communicate. This is what sets apart love and lust. And I think that it's better to, I think it's more appropriate to think of psychedelics as fostering this type of behavior. this altered state called love. And I could do a whole podcast on the different types of love and oxytocin neurons and the subtypes for different subtypes of love, but- Now you tell

us. That's what we would have done this podcast on. Okay, go on. Yeah, I mean, when I first started my lab, one of the things that we really worked on was trying to figure out the different subtypes of oxytocin neuron. And we discovered that there are the big oxytocin neurons that are important for what I call like mad love, like falling in love with your partner or falling in love with your baby, and then sort of platonic love that...

are encoded by oxytocin neurons that are smaller, release smaller amounts of this prosocial peptide oxytocin, and are really important for things like being able to form communal attachments, knowing who to let into your friend group and who not, and also encodes love in a much more guarded and nuanced way, and it seems that those two types of oxytocin neurons have evolved in parallel in species that do more or less of this hair-bonding type of love. You know, it's a rock star. B, what is it? And what are its origins? Yeah, so Ibogaine is a psychedelic that, you know, I can divide up psychedelics into different categories based on their acute subjective effects. Ibogaine is what is called an onyrogenic psychedelic, meaning onyrogen just means induces a dreamlike state. Yeah. And so.

you know, typically when people talk about, I begin trips, they're talking about, you know, this, Dreamlike state that feels very much like you're doing a history of your life in a slideshow or like a dreamscape People do life reviews people talk about how important those life reviews are for their ability to understand whatever psychological issues that they are having Ibogaine comes from a it's a it's a chemical that is synthesized from Iboga, which is a chemical that comes from the Tabernanthi Iboga plant. It's a tree and the Ibogaine is concentrated in the root and the bark of that tree. I found where? Where do you find that tree? It's found in Gabon mostly in Africa and it is traditionally used by the Buiti tribes and it's used dates back, you know, we depending on which sort of standard you use, and this is definitely not my field, but anthropological evidence suggests that it has been in continuous human use for at least 1,000 years, probably more than that. So it says, humans have been using Ibogaine for as long as, for example, psilocybin. Those two are probably the two oldest psychedelics that we have.

historical anthropological evidence that people have been using them. So yeah, and it's different from other psychedelics in so far as, you know, the trip is very, very long. And it has been especially interesting because it turns out that the veterans who are dealing with a huge amount of PTSD and depression and suicidality are trekking down to Mexico and other countries to use Ibogaine to help treat their PTSD and their addiction and their depression. Is that because it's viewed as an illegal substance here in the US? It is an illegal substance. It's a Schedule 1 psychedelic, and so it is not used therapeutically in the United States. But in Mexico, the legal status of Ibogaine is a little bit more unclear. It's basically not illegal, but it's not totally legal either. Who establishes these levels of drugs?

Controlled Substances Act of 1970 is what established psychedelics, including Ibogaine, as a Schedule I drug, and it is regulated by the DEA and the FDA. But because of this incredible therapeutic promise of Ibogaine, Americans for Ibogaine has done an amazing public awareness campaign and also legislative campaign to try and use some of the money that came to governments for reducing or addressing the opioid crisis. Starting first in Texas, what they have were able to do is get legislation passed that unlocks state-level money to do research on IVA gain for addressing, you know, sort of the opioid crisis in this country. And most recently, the Trump administration has acknowledge that this is a priority for them to try and get these medications, you know, research. I will say Ibogaine. I just want to be careful about Ibogaine, though, as powerful as it is and is sort of remarkably effective in the evidence, the anecdotal evidence and sort of this, you know, in the clinics in Mexico kind of evidence to suggest that it's useful for opioid use disorder. Ibogaine is a little bit. I mean, it has cardiotoxicity, meaning that if you take it...

and you have a pre-existing heart condition. It seems that you need to pair it with something that can stabilize the heart like magnesium in order to overcome this cardiotoxicity. So the FDA and the DEA are very interested in supporting this research. The National Institute of Drug Abuse actually just gave a group at Harvard, 11 million grant to work on a phase one safety study for Ibogaine. And that is gonna be really, really important to determine whether or not we can eventually use this medication clinically for opioid use disorder. That's great. So many of these rules are 50 years out

of date is the point. And somebody needs to get in there and at least review it every decade or so. I mean, there's gotta be some sensible way to keep you guys your labs active and vibrant in the effort to explore these connections.

That's right. And we are, you know, very excited about the possibility that, you know, MAPS, you know, you guys had Rick Doblin on the show a couple years ago. MAPS is an organization that spent the last 40 years working on trying to provide the clinical data to say, you know what, these drugs don't belong in schedule one because they do have therapeutic efficacy. And it's looking very hopeful that that approvals for MDMA and psilocybin may come through as early as this year. But we're all sort of waiting. I hope so. Wasn't marijuana a schedule one at some point? Am I remembering that correctly? Yeah, it was without a doubt. But that was because they were convinced that black people were using it. Right, right. And they were like, you know, we got to do something about these jazz musicians.

They're playing the notes all over the place. They must be high. Boy, you know it, little white kids will love jazz. Yeah, so if there's disproportionate usage in the black community, then you get to arrest them, but not the alcohol abusers that are in the white community. And then you have a disproportionate of people going to prison. Same thing with crack cocaine and regular cocaine. you go to jail for a lot longer, why? Because there's a disproportionate use of the black community. However, I will say this, what's great about what you're doing, what has to happen is people have got to understand that in general, no one wants to be addicted to drugs. That's right. Period. Like full stop, we gotta get over this whole notion in this nation.

and it's the pretty much like the only nation in the world that does this. Where we look at people whose brains have been hijacked by a substance, and we say the treatment for that is to throw your ass in jail. It is the dumbest frickin' thing in the world, and if we divorce ourselves from that philosophy, then it will free us to go ahead and explore ways that we might deal with the problem to help people and solve the problem at the same time. Chuck, for president. Yes. Before we sadly have to let you go, Gaul, I

must ask you, studying psychedelics, does it affect your relationship with reality? Thank you for asking that. With objective reality. Wow. Right. So true. I like that objective reality.

Yeah, I mean, I often end up on panels, especially within the psychedelics community, where, you know, there are four of us on stage and three of them say, oh, yeah, you know, psychedelics reveal that, you know, it pulls the curtain. And now I'm able to see the really real and I now I know that there's a reality beyond the physical world. I am a biologist and an evolutionary you know I see things in terms of evolution and biology and physicalism you know I started my career as an undergrad but my major was sponsored by the philosophy department and it was called comparative perspectives on the mind and I really sort of considered all different philosophical views on this and I have to say that for me the study of psychedelics is convinced me even more that we live in a physicalist reality and that the very fact that you can take a molecule, put it in your mouth and have that molecule essentially mimic another molecule that's in your brain already and then drastically alter your perception of reality and consciousness tells me that what we think of as consciousness is just the manifestation of two molecules interacting with each other, and that the universe is explainable even if we can't do it right now in terms of physical reality. Oh, you're great at a party, aren't you? God! Thanks for bringing me down, Gull! I know, I often get that. People are like, you're killing my dream, you know, but I just...

No, you're my hero. Neal's over here grinning from ear to ear. I've always said that the brain barely works enough to decode objective reality from that which is not. It barely succeeds at this. So to stir in some chemicals and then say, oh, now I can see it better, I don't get that. It could be an augmentation, Neil. It doesn't necessarily have to be a detraction. I mean, I think going back, though, to this notion of the noetic property, right? Like, it is a... I wouldn't necessarily frame it as the brain is barely working. Rather, I would say it's that we are interpreting the universe through our sensory modalities. And this... Everything that we see was evolved as a response to what was required to adapt and survive as an organ to not die to not die right and so to me there is potentially a reason why we.

would want to reopen critical periods. So I'll give you another example. We have known for many, many years that you can reopen critical periods by doing deprivation experiments. So in mice and humans, this looks like, for example, if you wanted to reopen the visual critical period, you can blindfold somebody for two weeks or blindfold an animal for two weeks, and then it'll reopen the visual critical period. And when you do that in humans, that experiment, it causes mystical experiences and hallucinations that are phenomenologically very similar to psychedelics. And we know that mystical traditions and religions have used this sort of deprivation type of experiment. for millennia to achieve mystical states, going and living in a cave for a month, or what the Zen Buddhists called beginner's mind. I mean, if you're looking for a neurobiological description of beginner's mind, critical period reopening is it. And so in my view, the sort of phenomenological overlap between psychedelic mediated critical period reopening and deprivation mediated critical period reopening suggests that what it feels like to be in this altered state of consciousness is just what it feels like to reopen critical periods. And I think that the reason that these things exist, like why does the brain need a mechanism for reopening critical periods? If it's so important that we close them, why would we ever want to be able to reopen them if we're an ancient person? Why would we do it with a psychedelic or a religious practice? And I think the reason is because Learned habits of memory are very useful, so long as your environment is stable and you're exposed to rules that you can follow and easily navigate the world following your sort of trained habits of thought and behavior. But imagine if something radically shifts in your environment, right? So if you're a mouse, if your whole mouse colony gets eaten by a a wolf or if you're a human and you move to a new culture where nobody speaks your language. These would all be circumstances where your brain would probably want to hit the reset button and say, you know what, the statistics of the world that I'm in right now are changed and I need to relearn for the environment that I now find myself in. And so I do think that these altered states are not really a broken brain so much as a brain experiencing a radical shift, kind of like if you and I were having this conversation and the video started flickering. We might not do anything with the flickering, but if it froze for any amount of time, we would hit the reset button, reset the internet, reset the browser and try and go again. And I

think that's kind of what the brain has evolved to be able to do when there are big shifts in the environment.

Can you repeat all that? You were breaking up. So I want to take this out. I want to take this out, if I may. So, Gull, I'm still stuck on your octopus, okay? And because we didn't... That was an hour and a half ago. I know it was an hour and a half, hour long ago. Let me just say that you already said, I did not know that octopi octopuses are asocial. So... Humans are generally not a social, we're very communal, generally, as a species for sure. So it seems to me, as much as we all love us some octopus, especially after the documentary, My Octopus Teacher, no one loved me some octopus, but as a next species to test, I don't see why you would glean any insights back into humans by treating octopuses, other than have a hilarious time watching them react. And the reason why I ask is, we already talked about not being able to teach an old dog new tricks for all the reasons you gave. Why not do this on dogs? And then demonstrate that you can teach an old dog new tricks and a dog is a mammal vertebrate. It's not a squishy invertebrate eight legged creature that squirts ink or whatever they do. Okay, so wouldn't dogs In this adage, can't teach an old dog new tricks, wouldn't that be a more useful other animal to test this on rather than an octopus? Dogs like meth. Instead? They told you. Dogs don't like MDMA. They like meth.

Well, I will tell you, you know, this is an old debate in neuroscience And in fact, I have to say my lab studies humans mice and octopuses And the reason for the octopus experiment is actually it's not our idea Jay-Z young who was sort of the great granddaddy of neuroscience meets ethology was the first to propose this idea that if you want to understand how to build complex behaviors from synapses molecules circuits you can certainly look at the human brain and then look at the most common closest relative. But another way to do it would be to look at an animal that is maximally different from humans and yet still capable of these complex behaviors. And what advantage of that is is that you get you don't have to get distracted by accidents of evolutionary history. So the best example of that is that, you know, an octopus doesn't have a cerebral cortex. It doesn't have a nucleus accumbens. It doesn't have a brain stem. It doesn't have an amygdala. None of those brain regions exist in an octopus. And yet when

you give them MDMA because the molecular similarity is there, you can induce these complex behaviors that are very similar.

And so that tells us that all of these studies out there that are doing like fMRI brain imaging and saying, oh, look, I gave a psychedelic or I gave whatever. I had people play Tetris or whatever experiment they're doing. And look, this part of the brain lit up. That kind of inference about the mechanistic relationship between those brain regions and those functions is a little bit misleading and that the true mechanism is really at the level of molecules and depending on the evolutionary history of the animal, they can use the sort of anatomical toolkit in whatever way makes sense for their evolutionary history, but you don't need a cortex to do it. It just would be really cool if you had an old dog and then you gave it your medicine and then it did a new trick. You could do that on The Today Show and then that would...

And you get all the funding you want after that. Well, I sort of, yeah, I mean, I sort of did that in mice and I'm hoping that if we can show this in humans, then we, you know, by doing the stroke experiment, then I think it will radically shift the way that we understand this mechanism and will really open up all kinds of other avenues of study and all kinds of other diseases. I mean, we are currently, for example, also interested in, does an octopus even have a critical period? Do they, you know, just to challenge this notion, like what are critical periods good for? You know, an octopus is able to regrow its limbs if you cut them off. Sometimes they cut them off themselves. There are some species of octopus that are so asocial that the male, in order to mate with the female, just dislodges his sperm-delivering hectocautilus arm so that he doesn't get eaten. And then he can regrow it. And so maybe they don't need critical periods in the same way that we do. And in fact, if they have some sort of novel innovation in their genetic code that enables them to do that, we want to know about it so that we can sort of co-opt their superpowers if having those superpowers doesn't come with some other, you know, risk or downfall that we don't want, right? And so we want to understand them. All I know is this. If you're ever with a woman who makes you have to dislodge your heptaconalist arm, you are with the wrong woman. You got a terrible taste of women. Okay? Come on, girl, I need that.

I do have to say working with octopuses is fun because they are unlike mice which are sort of you know the potato chips of the animal kingdom right everybody eats them so the tasty tasty snacks yeah so when they get scared though they you know they they do they are sort of obligate stoics, right? So they don't have much of a response. They freeze, they kind of cower in the corner. When you do something an octopus doesn't like, it will squirt you with water or ink and it will let you know. And I sort of, I like that they have attitude like that. It feels, you know, they're more similar to us. And that is why actually I think that there is such a...

instinct to anthropomorphize around octopuses and to sort of imagine that they're like as smart as us because they sort of occupy the same top predator role in their habitat. And so we have a tendency to see a lot of ourselves in them. But I would just ask everybody to exercise some caution about that kind of anthropomorphizing because the truth is, compared to a mouse, we are olfactory morons. Or even dogs, yeah. Kid me. But not to mention bears, please. But not only that, if an octopus wanted to imprison you, it would just need to put you in a room with a door that has three doorknobs. You would never get out You know an Aristotle thought that octopuses were the dumbest animal because they were easy to catch all you had to do is Dangle a piece of room, you know something shiny in front of them And they would reach out and explore it and then you just lift that shiny thing out of the water And so he thought they were just dumb And, you know, by that measure, you know, humans are pretty dumb too. You just dangle a couple of, you know, dollars in front of them and they'll go chasing. So, you know. Yeah, Aristotle got almost everything wrong in the physical science. Tell me about it. So, yeah. You have no hero worship about him from me. Yeah. Well, thank you. And we understand you were going to be with us in person, but the sea gods, Neptune and Poseidon got in the way. The storms closed Newark Airport.

And so thanks for joining us from your, you've been holed up in a hotel in Chicago. So thanks for joining us nonetheless in spite of those challenges. Thank you very much. It's been great. And how do we find your lab online? Is it, how do we track what you do? So my lab is at Dolan Lab, D-O-L-E-N-L-A-B dot O-R-G. Oh, cool. DolanLab.org. There you go. All right. Okay. Thanks for joining us once again. All right. So this has been another

installment of Star Talk, special edition. Chuck, thanks again for being here, Gary. Pleasure, my friend. Always having you guys here. Neil deGrasse Tyson, as always. Keep looking up.